

Circuit Breakers

Operating Principle · Arc Phenomenon · Arc Extinction · SF6 vs VCB

ELECTRICAL ENGINEERING | POWER SYSTEMS

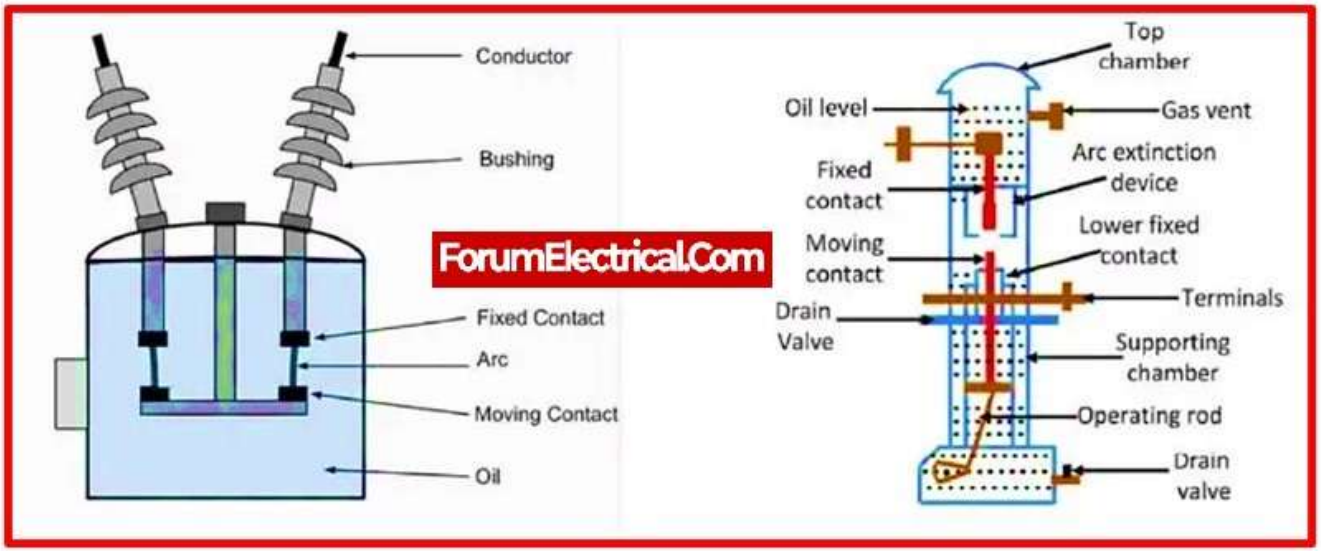


The Job in One Sentence

Protect equipment by interrupting fault current — fast enough that the power system survives.

A circuit breaker is the final, critical line of defence between a transient fault and catastrophic equipment failure. Every design decision flows from this single, non-negotiable requirement.

From Closed Contacts to a Trip



Normal Operation

Two conducting contacts carry load current continuously under normal system conditions.

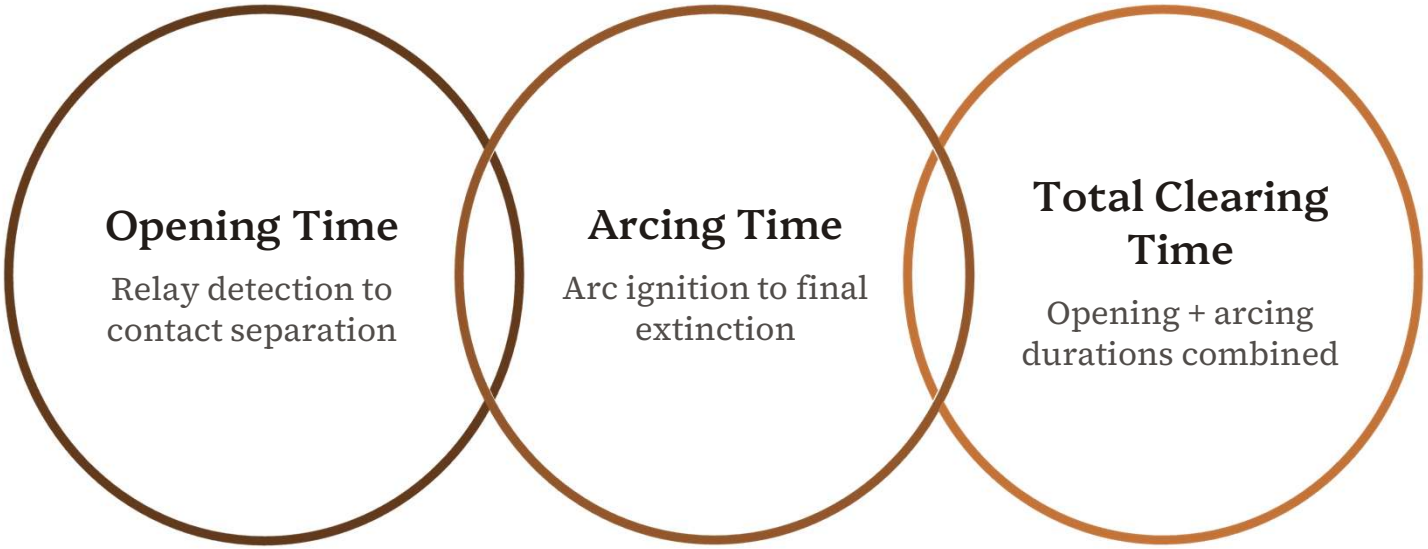
Fault Detected

A protective relay detects the fault and energises a **trip coil**, which mechanically drives the contacts apart.

Contact Separation

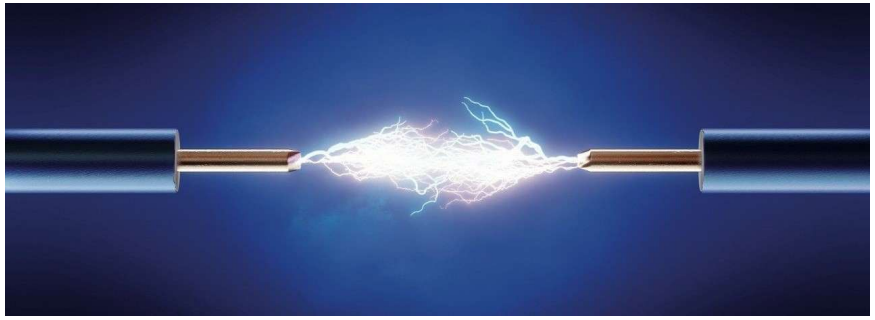
As the contacts part, the circuit is no longer metallic — the arc begins, and the interruption process takes over.

What "Clearing Time" Really Means



Minimising arcing time is therefore the dominant engineering challenge — the faster the arc is extinguished, the less damage the fault inflicts on the system.

The Instant Contacts Separate: Arc Ignition

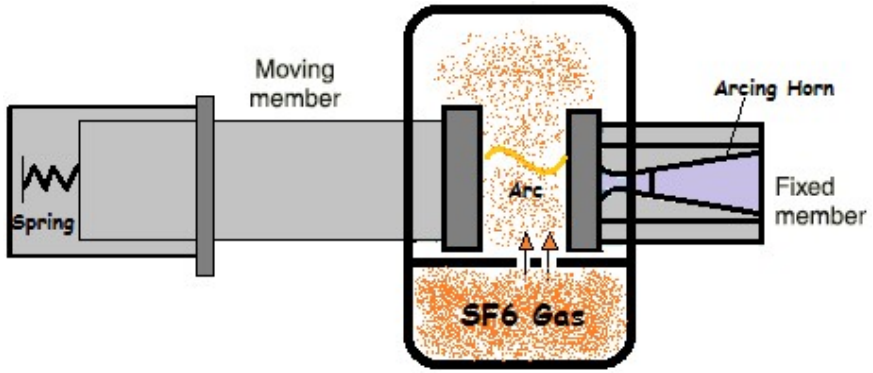


Contact Area Shrinks

As contacts begin to part, the conducting cross-section reduces rapidly, causing current density and temperature to spike.

Medium Ionises

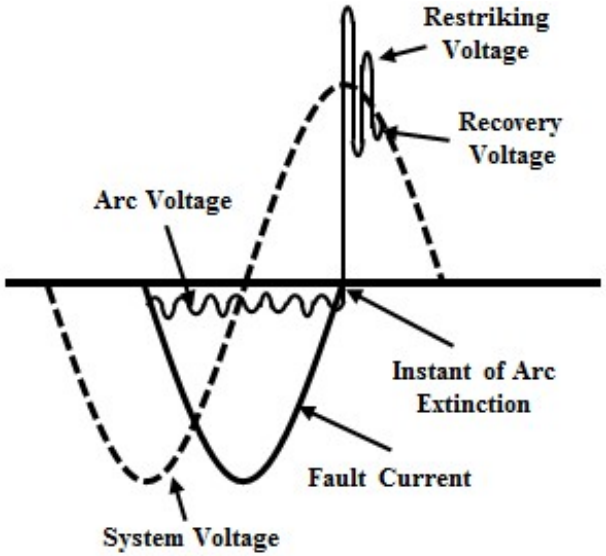
The intense heat ionises or vaporises the surrounding insulating medium, producing a conducting plasma channel.



Arc Forms

A luminous, high-temperature arc bridges the gap between the separating contacts, sustaining current flow.

Arc Voltage, Restriking & Recovery



Recovery Voltage, Restriking Voltage and Arc Voltage

Arc Voltage

A voltage drop appears across the arc while it conducts — this is the **arc voltage**.

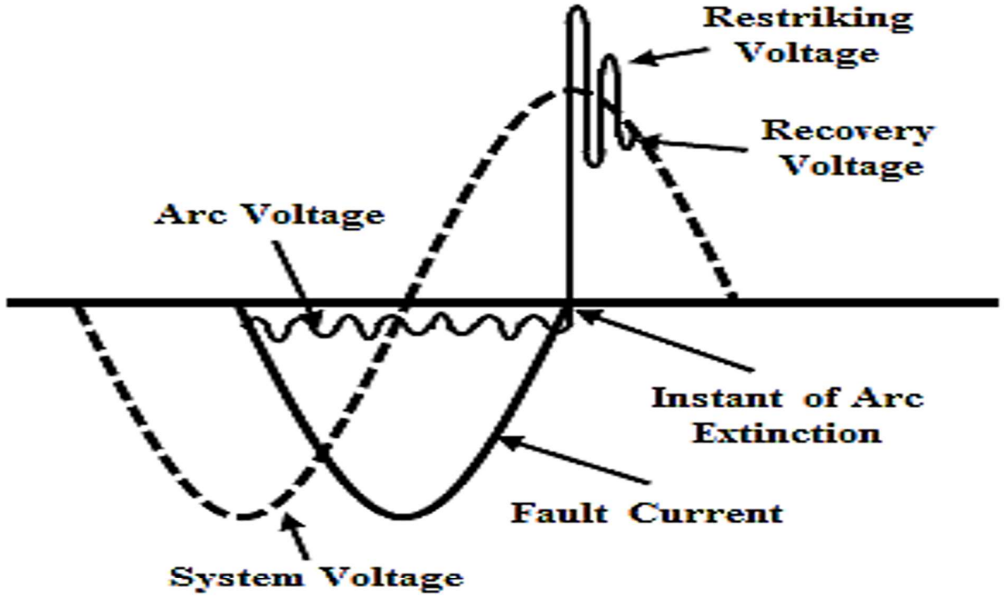
Restriking Voltage

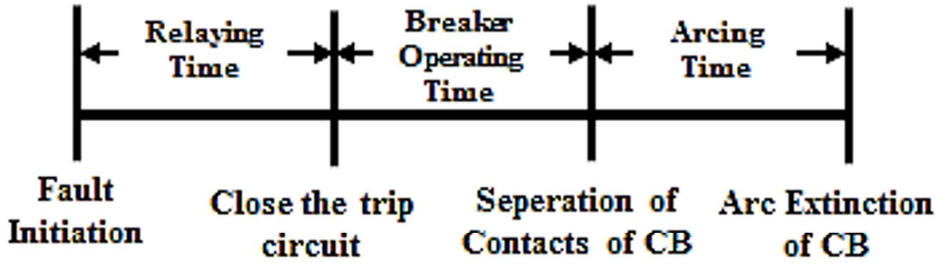
Near current zero, the system attempts to re-apply voltage across the gap. If dielectric strength has not recovered, the arc **restrikes**.

Recovery Voltage

After **final extinction**, the full recovery voltage appears across the now-open contacts.

Recovery Voltage, Restriking Voltage, Arc Voltage & Fault Clearing Time





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Why the Arc Wants to Live

Potential Difference

The voltage across the gap continuously drives current through the ionised channel, sustaining the arc.

Ionised Particles

Free electrons and ions between the contacts maintain conductivity — the arc is self-sustaining while these persist.

Arc Resistance Factors

Arc resistance is governed by the **degree of ionisation**, arc **length**, and arc **cross-sectional area**.

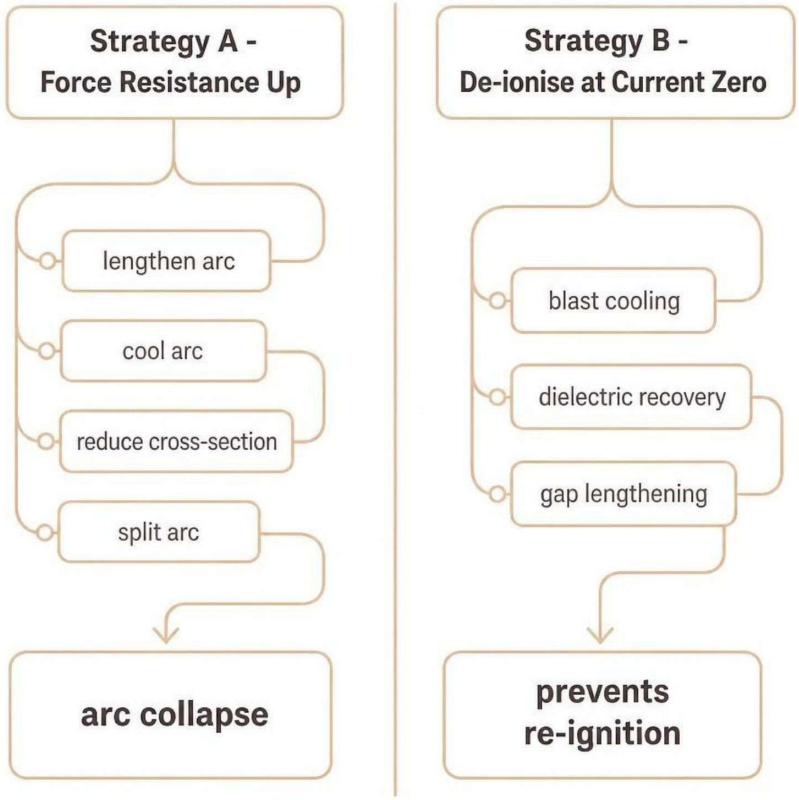
The Physics Leveraged into Design

Strategy A: Force Resistance Up

If arc resistance is driven high enough, the voltage required to sustain it exceeds what the system can provide — the arc collapses.

Strategy B: Die at Current Zero

AC current naturally passes through zero 100 times per second (50 Hz). Rapidly de-ionising the gap at this instant prevents re-ignition.



Both strategies exploit the same underlying physics — the arc cannot survive without ionised particles and sufficient driving voltage.

Two Arc Extinction Methods

High Resistance Method

Increase arc resistance progressively until the arc becomes unsustainable. Current is forced to zero before the natural AC zero crossing.

Used for: DC circuits & low-capacity AC

Low Resistance / Current-Zero Method

Maintain low arc resistance, then rapidly de-ionise the gap precisely at natural current zero to prevent restriking.

Used for: Modern HV AC circuit breakers

High Resistance Method: Starve the Arc



Lengthen the Arc

Increasing arc length raises resistance and makes the arc harder to sustain across a longer gap.



Cool the Arc

Removing heat reduces ionisation, increasing resistance. Arc cooling is achieved by contact geometry or surrounding medium.



Reduce Cross-Section

Constricting the arc column increases resistance per unit length, further impeding current flow.



Split the Arc

Arc splitter plates divide a single arc into a series of shorter arcs, each with its own voltage drop, multiplying total resistance.

High Resistance Method: The Energy Cost

Key Disadvantage

Because the arc is forced to carry current at elevated resistance for an extended period, **enormous energy is dissipated** within the arc itself — this creates thermal stress on contacts and surrounding components.

Where It Is Applied

- **DC circuits** — no natural current zero exists, so resistance must be forced up
- **Low-capacity AC circuits** — where arc energy remains manageable
- Fuses and certain low-voltage devices also use this principle

Low Resistance / Current-Zero Method

The dominant method for modern high-voltage AC circuit breakers. The arc is allowed to conduct — at **low resistance** — right up to the natural current zero of the AC cycle. At that precise instant, the gap is rapidly de-ionised so that the recovering system voltage cannot re-establish the arc.

Why Low Resistance?

Minimises energy dissipated in the arc during conduction, protecting contacts and the surrounding medium.

The Critical Window

De-ionisation must outpace dielectric recovery demand — if the gap re-ionises before the system voltage rises, the arc restrikes.

De-ionisation Tools

Lengthen the Gap

Increasing contact separation raises the voltage needed to restrikethe arc, buying time for de-ionisation.

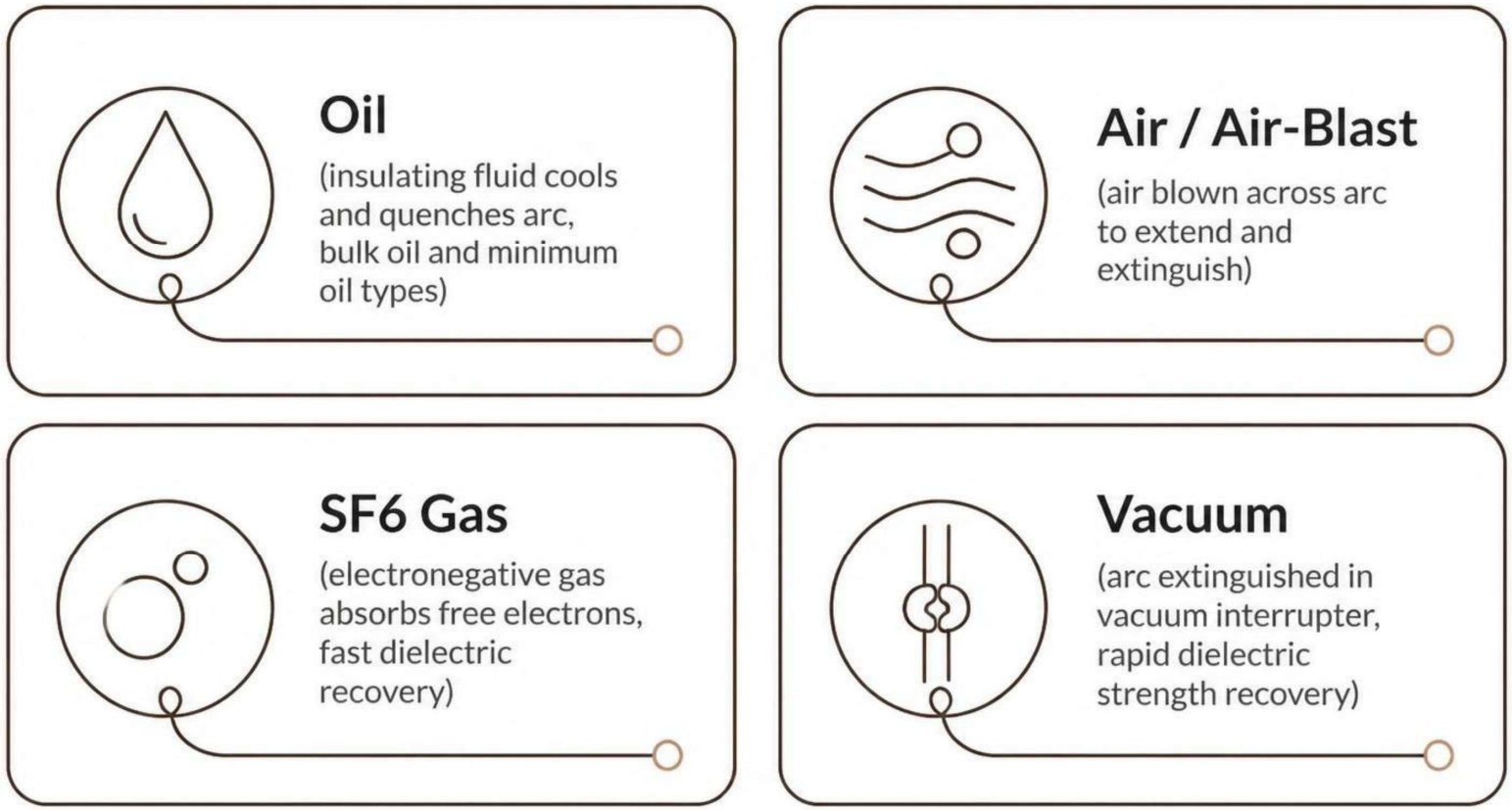
High-Pressure Blast Cooling

A directed gas or oil blast sweeps away ionised particles and cools the arc column rapidly and efficiently.

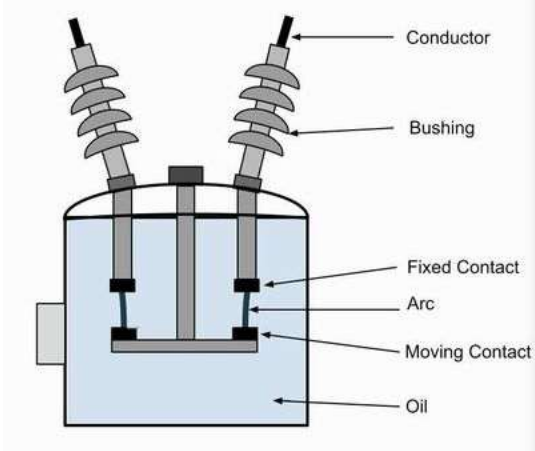
Dielectric Strength Recovery

The medium's insulating ability must be restored faster than the restriking voltage rises — achieved by pressure, gas properties, or vacuum.

Classifying Breakers by Arc Medium

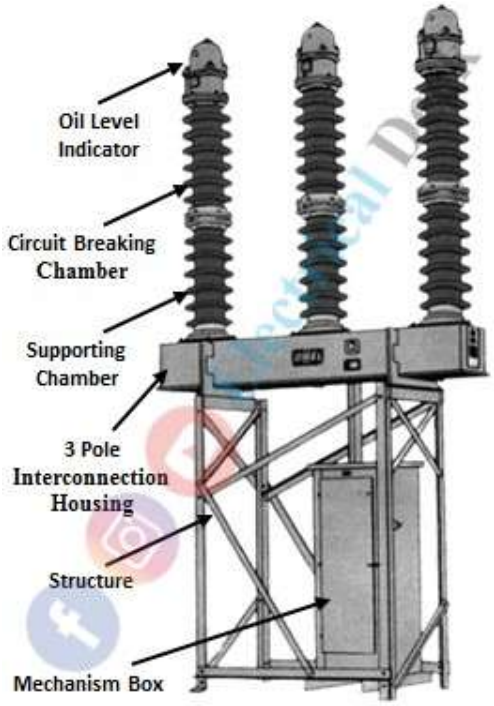






Oil Circuit Breakers

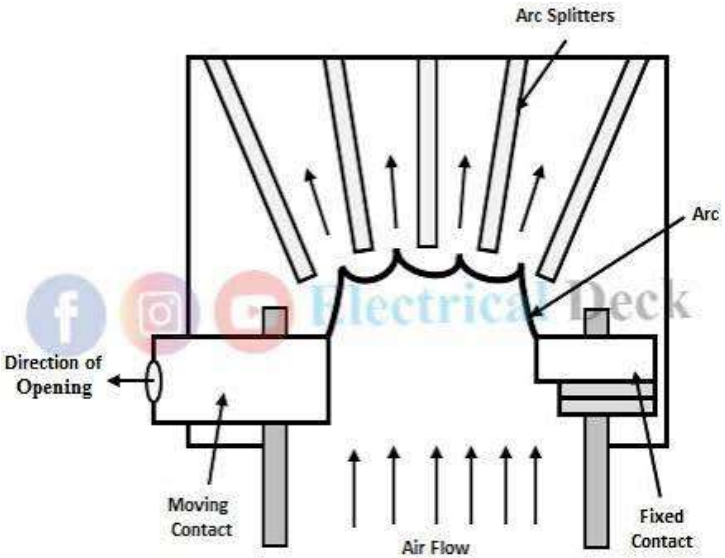
Oil surrounding the arc is vaporised and decomposed, producing hydrogen gas that cools and de-ionises. The oil also provides post-extinction insulation. Types include **bulk oil** and **minimum oil** breakers.



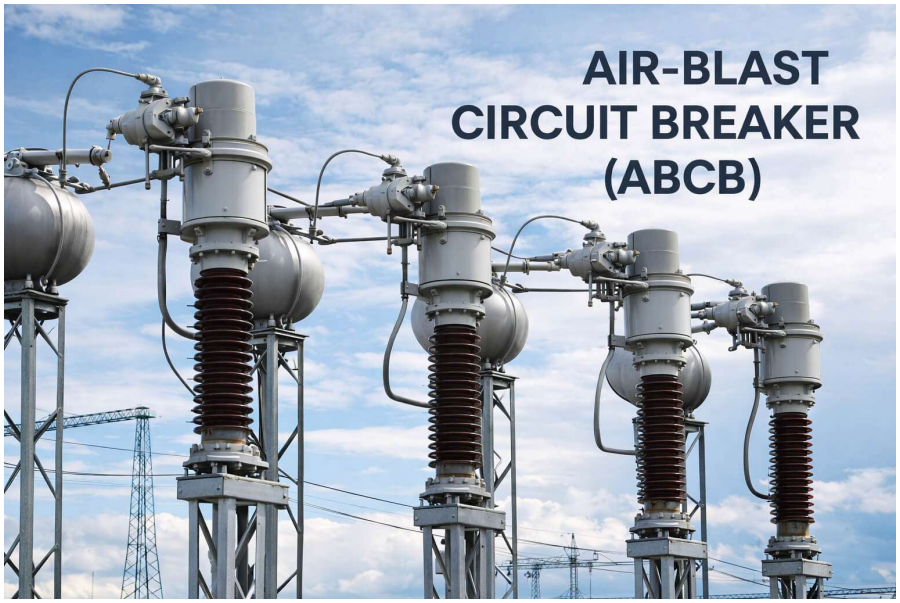
Air-Blast Circuit Breakers

Compressed air is directed across the arc at high velocity — extending, cooling, and ultimately extinguishing it.

Used extensively at extra-high-voltage levels before SF6 became dominant.



Cross Blast Air Circuit Breaker



SF6 Circuit Breakers: "Electrons Get Caught"

Electron Absorption

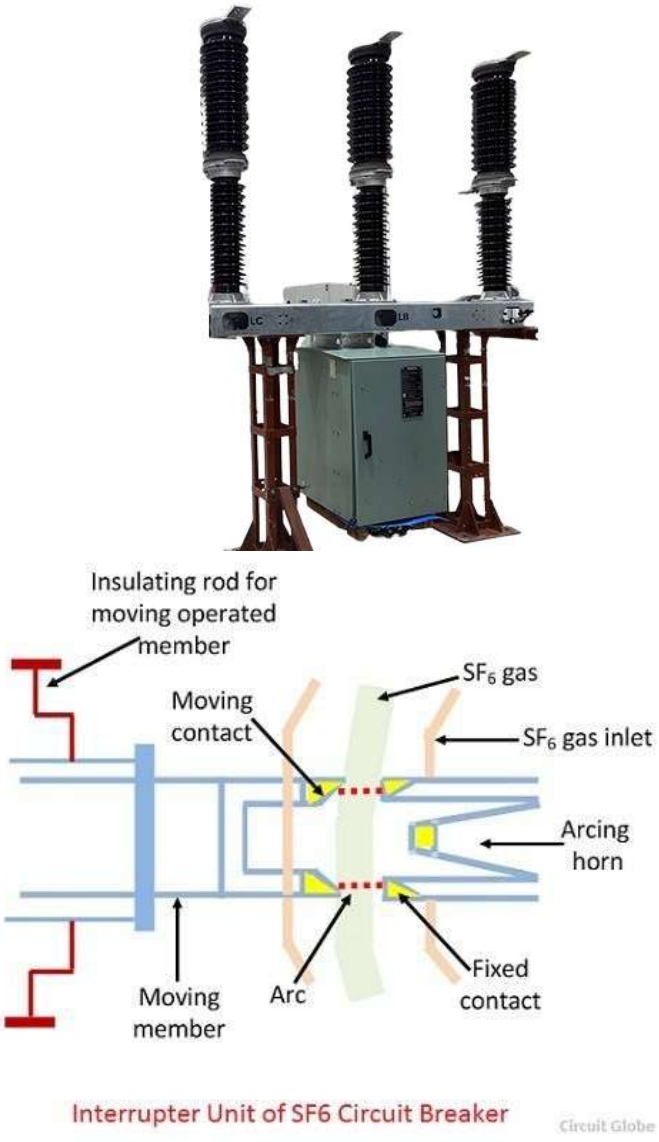
SF6 is **strongly electronegative** — it captures free electrons from the arc plasma, rapidly neutralising the ionised channel.

Short Arcing Time

Dielectric strength recovers extremely quickly after current zero, meaning the arc cannot restrike even under high recovery voltages.

High Interrupting Capability

SF6 breakers handle very large fault currents and are the preferred choice for **transmission-level voltages** (66 kV and above).



Interrupter Unit of SF6 Circuit Breaker

Circuit Globe

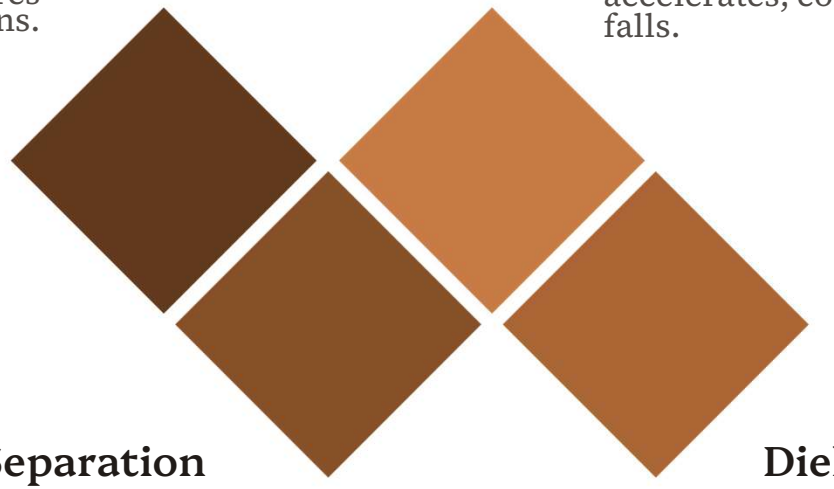
SF6 Arc Extinction: How It Happens

Arc in SF6

SF6 ionises and captures free electrons.

Current Zero

Electron capture accelerates; conductivity falls.



Contact Separation

Arc forms in SF6 atmosphere.

Dielectric Recovery

SF6 restores insulation; arc cannot restrike.

The combination of **electron affinity** and **high dielectric strength** (approximately 2.5× that of air at the same pressure) makes SF6 uniquely effective at both quenching the arc and holding off the recovery voltage.

Vacuum Circuit Breakers (VCB)

The Vacuum Interrupter

All switching takes place inside a hermetically sealed **vacuum interrupter bottle** at pressures of 10^{-4} to 10^{-6} torr — a medium with virtually no particles to ionise.

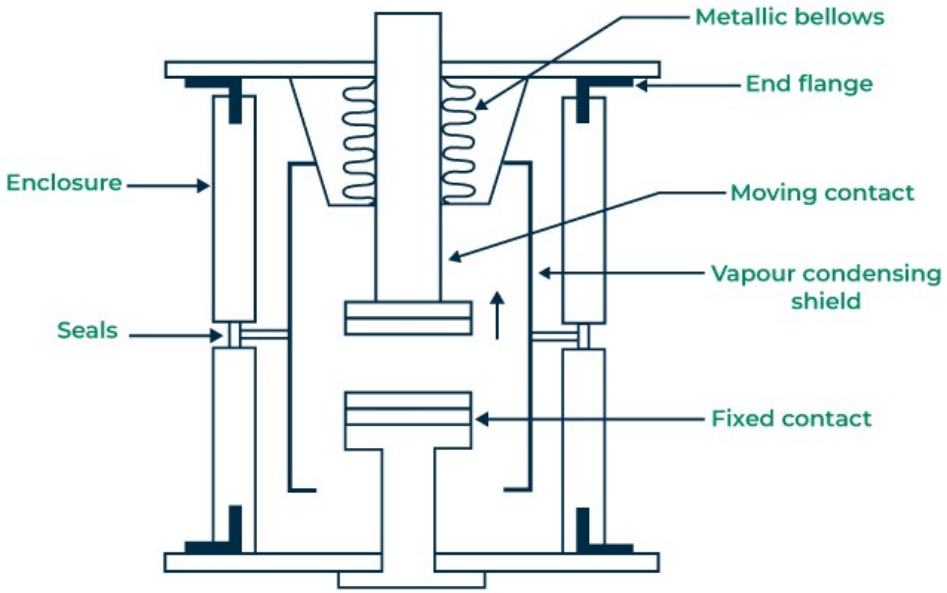
Arc Formation & Extinction

An arc *does* form on contact separation — sustained by **metal vapour** from the contacts themselves. However, at current zero, vapour condensation is nearly instantaneous and dielectric strength recovers **very rapidly**.

Key Advantages

- Compact and lightweight interrupter
- Low maintenance — contacts sealed for life
- Suitable for frequent switching operations
- No arc-quenching gas required




Metallic bellows
End flange
Enclosure
Moving contact
Vapour condensing shield
Seals
Fixed contact



Conclusion: Choosing SF6 vs VCB

SF6 Breakers

Best for: Transmission and sub-transmission voltages (66 kV+), high fault-current duties, and large gas-insulated switchgear (GIS) installations.

Watch for: Environmental obligations — gas monitoring, leak prevention, and emerging regulatory restrictions on new installations.

Vacuum Circuit Breakers

Best for: Medium-voltage networks (up to ~36 kV), industrial and commercial switchgear, frequent switching, and environments where compact, low-maintenance design is essential.

Advantage: No harmful gas, long service life, and very fast dielectric recovery at current zero.

Both technologies rely on the same fundamental goal: de-ionise the gap faster than recovery voltage can restrike the arc. The medium chosen determines how that race is won.